MUHAMMAD HUSNAIN MUBARIK

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RESEARCH INTERESTS

Computer Architecture, Hardware Acceleration of Machine Learning Applications, Hardware for Graphics (Conventional, Neural and Generative), Hardware Acceleration of Generative AI, Printed Computer Systems, Emerging Technologies and Applications.

EDUCATION

University of Illinois at Urbana Champaign (UIUC), USA. *Jan 2019 - May 2024*
PhD - Electrical and Computer Engineering (ECE)
Advisor: Prof. Rakesh Kumar

National University of Sciences and Technology (NUST), Pakistan. *Sep 2013 - July 2017*
Bachelors of Science in Electrical Engineering.
CGPA: 3.95/4.00, **summa cum laude**.

PUBLICATION SUMMARY AND IMPACT

- 6× (ISCA, MICRO), 1× (DATE)
- 1× ISCA Retrospective (1996-2020) Selection
- 2× IEEE Top Picks Honorable Mention
- Dan Vivoli Endowed Fellowship
- Served as a program committee (PC) member at HPCA, ISCA, ASPLOS, MLSys, and DAC.
- Emerging Vision and Graphics Systems and Architectures (EVGA), workshop organization, colocated with ISCA 2024.

RESEARCH AND WORK EXPERIENCE

Advanced Rendering Research (ARR), AMD, USA.

- Senior Member of Technical Staff (SMTS) Software Development Eng. *June 2024 - **Present***
  Full-time, Manager: Bruno Stefanizzi
- From Bit-Position Sensitivity to Unequal Error Protection for DNN Inference Memory (**MICRO-59th, 2026**)
  - Characterized per-bit-position fault sensitivity of DNN inference across 16 models, 6 modalities, and three floating-point formats (FP16, BF16, FP32).
  - Identified a sharp bit-sensitivity threshold (Xsafe) below which fraction LSBs can remain unprotected, showing uniform SECDED (12.5% overhead) is unnecessarily conservative; derived per-data-type floors and workload-aware tiers that raise ECC savings to 37.5–62.5% without retraining.
  - Designed an Unequal Error Protection (UEP) codec with per-cacheline data-type tags and a dual-partition SRAM architecture that grades protection by bit significance (full SECDED on sign/exponent, single-error correction on high-order mantissa, none on tolerant LSBs).

– Validated selective protection under contiguous 2- and 3-bit upsets across 870+ fault-injection runs; the codec reduces ECC area by 27.8% versus uniform SECDED and dual-voltage operation lowers BF16 read energy by ∼17%.

Graduate Research Intern, Intel, USA.

- **Emerging Visual-AI Systems Research Lab** *February 2024 - May 2024*
  Part-time (20 hrs), Managers: Nilesh Jain, Ravishankar Iyer.
- **Emerging Visual-AI Systems Research Lab** *September 2023 - February 2024*
  Part-time (10 hrs), Managers: Nilesh Jain, Ravishankar Iyer.
- **Graphics Research Organization (GRO)** *May 2023 - August 2023*
  Full-time, Managers: Tobias Zirr, Anton Sochenov, Anton Kaplanyan.
- **Next-Gen Architecture (NGA)** *September 2022 - May 2023*
  Part-time, Manager: Maxim Kazakov.
- **Graphics Research Organization (GRO)** *June 2022 - August 2022*
  Full-time, Managers: Rama Harihara, Anton Kaplanyan.
- **Cloud Systems Research Lab (CSR)** *December 2021 - May 2022*
  Part-time, Managers: Nilesh Jain, Ravishankar Iyer.
- **Heterogeneous Platforms Lab (HPL)** *May 2021 - August 2021*
  Full-time, Manager: Tanay Karnik.

Coordinated Science Laboratory (CSL), UIUC, USA *January 2019 - May 2024*
Research Assistant
Advisor: Prof. Rakesh Kumar

- Hardware Acceleration of Latent Diffusion Models (Undersubmission)
- ML accelerators for accelerating gaming and vision applications (Ongoing)
- Hardware Acceleration of Neural Graphics (**ISCA-50th, 2023**)
  – Neural graphics as a potent replacement for traditional rendering algorithms.
  – In this paper we addressed the question does NG need HW support?
  – Disparity between desired performance and the current state of the art (1.5x-55x).
  – Identification of input encoding and MLP kernels as performance bottlenecks.
  – Proposal of NGPC – a hardware solution to accelerate input encoding and MLP.
  – The NGPC achieves up to 58X end-to-end application-level performance improvement.
  – NGPC enables rendering of 4k Ultra HD resolution frames at 30 FPS for NeRF and 8k Ultra HD frames at 120 FPS for all other neural graphics applications.
- Understanding Interactions Between Chip Architecture and Uncertainties in Semiconductor Supply and Demand (Under Submission - Preprint available on arXiv.)
- Exploiting Short Application Lifetimes for Low Cost Hardware Encryption in Flexible Electronics (**DATE 2023**)
- Rethinking Programmable Earable Processors (**ISCA-49th, 2022**)
  – We first explore the hardware requirements for earable computing platforms like earphones, hearing aids, and smart glasses.

- we propose EarBench, a suite of earable applications, to analyze performance gaps between earable computational needs and the capabilities of existing microprocessors.
- Analysis reveals a performance gap of 13.54x-3.97x on average, with more complex microprocessors being energy-inefficient for earable applications.
- EarBench applications are found to be dominated by a few DSP and ML-based kernels with significant computational similarity.
- The proposed solution is SpEaC, a reconfigurable spatial architecture optimized for energy-efficient execution of earable kernels at the cost of generality.
- SpEaC outperforms modeled programmable cores (M4, M7, A53, HiFi4 DSP) by up to 99.3x, and offers substantial energy efficiency benefits, outperforming a low power Mali T628 MP6 GPU by 15.7x - 1087x.

- Model-specific Design of Deeply-Embedded Tiny Neural Network Accelerators (Ongoing).
  - Current edge NN accelerator hardware designs tend to be model-agnostic and are, therefore, over designed and expensive for a specific model.
  - For a suite of tiny neural networks, we show that customizing hardware to model structure can provide 101× benefits in terms of inference throughput and 69× benefits in terms of energy efficiency over Eyeriss. Additional 4.3× and 2.4× benefits are achievable in terms of area and energy efficiency respectively if model data is also known during design time.
- Printed Machine Learning Classifiers (**MICRO-53rd, 2020**)
  - Printed electronics meet cost, conformity, and non-toxicity needs unfulfilled by silicon-based systems in many applications.
  - We first explore the accuracy and cost of classification algorithms for two printed technologies, EGT and CNT-TFT.
  - Our EGT-bespoke Decision Tree and SVM classifiers exhibit 4x, 75x, 48x and 1.4x, 12.7x, 12.8x improvements in delay, power, and area, respectively, over conventional systems.
  - Our lookup-based classifiers enhance area and power efficiency by 1.93x and 1.65x with a 50% delay overhead for EGT Decision Trees. SVMs present a 40% delay cut and 8% and 1% area and power improvements.
  - EGT analog Decision Trees and SVMs surpass their digital equivalents in area and power by 437x, 27x and 490x, 1212x, respectively, but are 1.63x slower.
  - We presented several functional printed prototypes.
- Printed Microprocessors (**ISCA-47th, 2020.)**
  - We first explore the design space for microprocessors implemented in printing technologies, also devising standard cell libraries for these technologies.
  - Our design space exploration of printed microprocessor architectures across various parameters reveals our best cores surpass pre-existing ones by at least an order of magnitude in power and area.
  - Introducing printed-specific optimizations and Program-specific ISA (TP-ISA), we achieve up to 4.18x and 1.93x improvements in power and area, respectively. A Cross point-based instruction ROM outperforms a RAM-based design by 5.77x, 16.8x, and 2.42x in power, area, and delay, respectively.

PUBLICATIONS

1 **M. Mubarik**, K. Mohan Kumar, P. A. Pena, K. Varadarajan, K. Tyagi, "From Bit-Position Sensitivity to Unequal Error Protection for DNN Inference Memory," **MICRO-59th, 2026.**

2 N. Bleier, **M. Mubarik**, G. Swenson, R. Kumar "Space Microdatacenters" **MICRO-56th, 2023, IEEE Micro Top Picks - Honorable Mention 2024.**

3 **M. Mubarik**, R. Kanungo, T. Zirr, R.Kumar "Hardware Acceleration of Neural Graphics" **ISCA-50th, 2023.**

4 N. Bleier, **M. Mubarik**, S. Balaji, F. Rodriguez, A. Sou, S. White and R. Kumar, "Exploiting Short Application Lifetimes for Low Cost Hardware Encryption in Flexible Electronics," **DATE 2023.**

5 N. Bleier, **M. Mubarik**, S. Chakraborty, S. Kishore, R. Kumar, "Rethinking Programmable Earable Processors," **ISCA-49th, 2022.**

6 **M. Mubarik**, D. Weller, N. Bleier, M. Tomei, J. Aghassi-Hagmann, M. Tahoori, R. Kumar, "Printed Machine Learning Classifiers," **MICRO-53rd 2020, IEEE Micro Top Picks - Honorable Mention 2021.**

7 N. Bleier*, **M. Mubarik***, F. Rasheed*, J. Aghassi-Hagmann, M. Tahoori, R. Kumar, "Printed Microprocessors," **ISCA-47th, 2020.** [* Co-First Authors, Listed in Alphabetical Order] **Selected for the retrospective of the years 1996 through 2020 on the 50th anniversary of ISCA.**

8 R. Kanungo, S. Siva, N. Bleier, **M. Mubarik**, L. Varshney, R. Kumar "Understanding Interactions Between Chip Architecture and Uncertainties in Semiconductor Supply and Demand" - Under Submission, preprint arXiv:2305.11059, 2023.

TALKS

- Delivered a crash course on diffusion models at AMD, December 2024.
- Delivered a guest lecture on "Neural Rendering" at AMD, June 2024.
- Delivered a guest lecture on "Tensor Processing Units" at Intel Labs, April 2024.
- Delivered a guest lecture on "Accelerators/Specialization/Emerging Architectures" in ECE 411, Spring 2024.
- Delivered a guest lecture on "GPU Architecture and Programming" in ECE 411, Spring 2024.
- Gave a talk titled "Hardware Acceleration Opportunities in Generative AI Based Graphics" at Intel [2024].
- Presented "Hardware Acceleration of Neural Graphics" paper at ISCA-50th, 2023.
- Gave a talk on "Neural Graphics: An Architecture's Perspective" at ECE498SJP: Accelerator Architectures class at UIUC [2023].
- Gave a talk titled "Hardware Acceleration Opportunities in Neural Radiance Fields" at Intel [2022].
- Presented "Printed Machine Learning Classifiers" paper at MICRO-53rd, 2020.
- Gave a talk on "Hardware for Deep Learning" at ECE511: Advanced Computer Architecture class at UIUC [2021].

SELECT ACADEMIC ACHIEVEMENTS AND AWARDS

- **Dan Vivoli Endowed Fellowship (NVIDIA) for the 2024-2025 academic year.**
- **IEEE Micro Top Picks - Honorable Mention, 2024.**

- Our paper ”Space Microdatacenters” (MICRO-56th, 2023) ranks among the Top 24 papers published in computer architecture venues in 2023.

- **ISCA-50th 25 year retrospective (1996-2020), 2023.**
  - Selected for the retrospective of the years 1996 through 2020 on the 50th anniversary of ISCA for our paper ”Printed Microprocessors” (ISCA-47th, 2020).
- **Machine Learning and Systems Rising Star, 2023.**
- **ECE-UIUC travel grant for attending ISCA-2023.**
- **IEEE Micro Top Picks - Honorable Mention, 2021.**
  - Our paper ”Printed Machine Learning Classifiers” (MICRO-53rd, 2020) ranks among the Top 24 papers published in computer architecture venues in 2020.
- **Prime Minister’s Silver Medal**, NUST-CEME, 2018.
  - 2nd highest CGPA (3.95/4.00) in the Electrical Engineering Department at NUST-CEME.
- Travel award for EECamp at KAIST, South Korea, 2018.
- NUST GPA scholarships 2013-2017.
- High Achievers Award, NUST-CEME, 2015.
- Golden Star Award, NUST-CEME, 2015.

TECHNICAL SKILLS

- Programming languages: C++, Python, Verilog, SystemVerilog, HIP, CUDA, GLSL, Scala, SystemC, Tcl, Bash, Makefile
- Software: PyTorch, TensorFlow, TensorRT, Onnx-Runtime, Caffe, Keras, Synopsys-VCS, MATLAB, LaTeX, Vim
- Frameworks: CUDA, Nvidia Nsight (Compute/Systems/Graphics), Vulkan, OpenGL, CHISEL, gem5.

PROFESSIONAL SERVICES AND ACTIVITIES

- Conference Roles:
  - Program Committee (PC) Member:
    - International Symposium on Computer Architecture (ISCA) – 2025, 2026.
    - Design Automation Conference (DAC) – 2025, 2026.
    - Conference on Machine Learning and Systems (MLSys) – 2025, 2026.
    - Architectural Support for Programming Languages and Operating Systems (ASPLOS) – 2026.
    - International Symposium on High-Performance Computer Architecture (HPCA) – 2025.
    - Emerging Vision and Graphics Systems and Architectures (EVGA), co-located with ISCA – 2024.
  - Artifact Evaluation Committee (AEC) Member: ASPLOS – 2025.
  - Reviewer: ACM Transactions on Architecture and Code Optimization (TACO).
- Workshop Organization:

- ML and Systems Rising Stars, 2024 - Present.
- Emerging Vision and Graphics Systems and Architectures (EVGA), co-located with the 51st International Symposium on Computer Architecture (ISCA), 2024.

TEACHING AND MENTORING EXPERIENCE

- Mentored student projects in ECE 508 (Manycore Parallel Algorithms, Spring 2023) at UIUC.
- Frank Cai — 4th-year Undergraduate Student
  Project titled ”Hardware Acceleration Opportunities for Speculative Decoding to Accelerate LLMs.” Secured admission at UIUC's Graduate College.
- Ramakrishna Kanungo — 2nd year Masters Student
  Hardware Accelerator for Neural Radiance Field Inference. Secured industry job in load-store unit design.
- Abigail Wezelis — 2nd year Masters Student
  Design space exploration of data specific architectures for accelerating DNNs. Secured industry job in system design.
- Srijan Chakraborty — 2nd year Masters Student
  Synthesis and performance/area/power analysis of very large RTL designs.

SELECT COURSES

- ECE498SJP: Accelerator Architectures (Sanjay Patel) - Spring 2023.
- CS534: Advanced Topics in Computer Architecture (Sarita Adve) - Spring 2022.
- ECE598JH: Advanced Memory and Storage Systems (Jian Huang) - Fall 2021.
- CS598DHK: 3D Vision (Derek Hoiem) - Fall 2021.
- CS498ME Architecture for Mobile and Edge Computing (Saugata Ghose) - Spring 2021.
- CS598LCE Languages and Compilers for Edge Computing (Vikram S. Adve) - Spring 2021.
- ECE598NSG: Deep Learning in Hardware (Naresh Shanbhag) - Fall 2020.
- ECE524/CS563: Advanced Computer Security (Adam Bates) - Spring 2020.
- CS598SVA: App-Cust Heterogeneous Systems (Sarita Adve) - Spring 2020.
- CS598JT: Energy Efficient Computer Architecture (Josep Torrellas) - Fall 2019.
- ECE511: Advanced Computer Architecture (Rakesh Kumar) - Spring 2019.
- CS533: Parallel Computer Architecture (Josep Torrellas) - Spring 2019.